Cloud Data Platform

Technical Datasheet

AWS + Databricks Lakehouse
End-to-End Real Data Pipeline

Version 1.2
Phase: P0 - Decisions and Project Bootstrap
Status: DECIDED / SELECTED / PENDING MANUAL VERIFICATION
Updated 2026-10-13: ADR-007 (Supabase replaces RDS); ADR-008 proposed (optional streaming path)

A. PROJECT OVERVIEW

The Cloud Data Platform is an end-to-end, portfolio-scale data engineering project that demonstrates production-grade practices for ingesting real external data, transforming it through a medallion architecture, applying data quality controls, and exposing business-oriented analytics.

The platform runs on AWS and Databricks. It ingests from two public REST APIs (World Bank Indicators, FRED Economic Series) and one operational database (Supabase PostgreSQL via Fivetran). Data flows through immutable S3 raw storage, Bronze/Raw Fivetran layers, dbt-managed Silver and Gold transformations, and is governed by Unity Catalog. Orchestration uses Amazon MWAA (Airflow) as an ephemeral environment created and destroyed per work window. Infrastructure is codified in Terraform with GitHub Actions CI/CD using OIDC authentication.

An optional, time-boxed streaming path (P8b) is proposed in ADR-008. It is not part of the approved scope and no implementation may start until ADR-008 is Accepted.

B. ARCHITECTURE

Architecture Diagram

See docs/project-architecture.png for the visual architecture reference. The diagram shows two core ingestion paths converging at dbt transformations, plus an optional proposed streaming path:

Path A - API Ingestion:

- World Bank API -> Python Client -> MWAA/Airflow -> S3 Immutable Raw -> Databricks Bronze
- FRED API -> Python Client -> MWAA/Airflow -> S3 Immutable Raw -> Databricks Bronze

Path B - Database Ingestion:

- Supabase PostgreSQL (macro_watchlist_db, schema macro_watchlist) -> Fivetran (PostgreSQL connector, Query-Based sync through the Supabase session pooler) -> Databricks (schema: raw_fivetran, prefix to be confirmed in P8)

Path C - Streaming (optional, proposed in ADR-008, not selected):

- Public streaming feed (WebSocket/SSE) -> local time-boxed producer (reconnect, validation, PutRecords) -> Kinesis Data Streams (1 shard, 24h retention; ephemeral stack, aws/streaming Terraform root, applied/destroyed manually, never by CI) -> Databricks streaming job -> bronze.stream_<source> -> dbt incremental Silver -> Gold
- Invalid messages -> S3 raw bucket prefix quarantine/stream/<source>/. Checkpoints in a non-raw location. Streaming freshness shown in the data-quality dashboard.

Shared Transformation Layer:

- Bronze + raw_fivetran (prefix TBC in P8) -> dbt Silver (staging) -> dbt Gold (marts) -> Databricks AI/BI

Quarantine (Hybrid per ADR-002):

- Structural failures (malformed JSON, schema mismatch) -> S3 quarantine prefix
- Semantic failures (business rules, PK/FK, domain values) -> Unity Catalog quarantine schema

C. DATA SOURCES

World Bank Indicators (API 1)

Property	Value
Source Name	world_bank_indicators
Type	Public REST API
Provider	World Bank
Base URL	https://api.worldbank.org/v2
Authentication	None
Pagination	page + per_page (max 500)
Response Format	JSON
Key Endpoints	/country/{country}/indicator/{indicator}
Initial Indicators	NY.GDP.MKTP.KD.ZG, FP.CPI.TOTL.ZG, SL.UEM.TOTL.ZS
Expected Volume	~6,000 observations/run (3 ind x 200 countries x 10 yrs)
Data Grain	One observation per country + indicator + year
Raw S3 Path	s3://<bucket>/api/world_bank_indicators/ingest_date=YYYY-MM-DD/
Bronze Destination	<catalog>.bronze.world_bank_indicators
Status	Selected

FRED Economic Series (API 2)

Property	Value
Source Name	fred_economic_series
Type	Public REST API
Provider	Federal Reserve Bank of St. Louis
Base URL	https://api.stlouisfed.org/fred
Authentication	API Key (via Secrets Manager)
Pagination	limit + offset
Response Format	JSON
Key Endpoints	/series/observations
Initial Series	CPIAUCSL, UNRATE, FEDFUNDS
Expected Volume	~780 observations/run (3 series x 5 yrs monthly)
Data Grain	One observation per series + observation date
Raw S3 Path	s3://<bucket>/api/fred_economic_series/ingest_date=YYYY-MM-DD/
Bronze Destination	<catalog>.bronze.fred_economic_series
Status	Selected

Supabase PostgreSQL via Fivetran (Operational DB)

Property	Value
Source Name	macro_watchlist_db
Type	Operational PostgreSQL Database
Provider	Supabase
Database	Supabase PostgreSQL, schema macro_watchlist (via session pooler)
Fivetran Connector	PostgreSQL
Authentication	DB credentials (Secrets Manager secret cloud-data-platform/macro-watchlist-db; container only, no values in Git)
Sync Frequency	Daily / On-demand via Airflow
Expected Volume	< 1,000 rows total
Data Grain	One row per entity (country, indicator, alert rule)
Destination	dwh_dev.raw_fivetran.<table> (Fivetran prefix to be confirmed in P8)
Incremental Strategy	Query-Based sync; exact cursor (e.g. xmin) to be confirmed in P8 via the session pooler
Tables	watchlist_country, watchlist_indicator, alert_rule
Setup	Supabase project created manually, outside AWS Terraform; the AWS persistent stack only holds the secret container
Decision	ADR-007 (supersedes ADR-005, Amazon RDS)
Status	Selected

Public Streaming Feed (optional, proposed)

Property	Value
Source Name	Not selected
Type	Public WebSocket / SSE feed
Path	Local producer -> Kinesis -> Databricks streaming job -> bronze.stream_<source> (ADR-008)
Business Question	BQ-004 not defined (open)
Status	Not selected (ADR-008 Proposed; no implementation until Accepted)

D. INITIAL DATA CONTRACTS

World Bank Indicators

Indicator Code	Name	Unit	Frequency	Historical Range
NY.GDP.MKTP.KD.ZG	GDP Growth (annual %)	Percent	Annual	10 years
FP.CPI.TOTL.ZG	Inflation, consumer prices (annual %)	Percent	Annual	10 years
SL.UEM.TOTL.ZS	Unemployment, total (% of labor force)	Percent	Annual	10 years

FRED Economic Series

Series ID	Name	Unit	Frequency	Historical Range
CPIAUCSL	Consumer Price Index for All Urban Consumers	Index 1982-84=100	Monthly	5 years
UNRATE	Civilian Unemployment Rate	Percent	Monthly	5 years
FEDFUNDS	Federal Funds Effective Rate	Percent	Monthly	5 years

Supabase PostgreSQL Tables (macro_watchlist_db)

Table	Key Fields	Purpose
watchlist_country	country_id, country_code, country_name, region, active, added_at	Countries to monitor
watchlist_indicator	indicator_id, source_system, source_key, indicator_name, unit, frequency, active	Indicators to track (WB/FRED)
alert_rule	alert_rule_id, country_code, indicator_id, threshold, direction, active, created_at	Alert thresholds per country/indicator

E. DATA LAYERS

The platform implements a medallion architecture with additional quarantine and snapshot layers:

- **S3 Raw/Landing:** Immutable source payloads. API ingestions create new objects per ingestion date; never overwrite. Partitioned by ingest_date=YYYY-MM-DD.
- **Bronze:** Data close to source plus technical metadata: _batch_id, _ingested_at, _source_file, _source_system, _batch_date. Loaded via COPY INTO (ADR-001).
- **raw_fivetran:** Fivetran-managed landing area for the operational database (Supabase). Direct write to Databricks, no S3 intermediary (D-002). Schema prefix to be confirmed in P8.
- **Silver:** Parsing, types, deduplication, normalization, business rules. Implemented as dbt staging models.
- **Gold:** Analytical dimensions and facts in dbt marts. Stable business definitions for dashboards.
- **Quarantine:** Rejected records with diagnostic metadata. Structural -> S3 quarantine; Semantic -> Unity Catalog quarantine schema (ADR-002).
- **Snapshots:** dbt snapshot history using SCD2 where historical tracking is required (e.g., watchlist changes).
- **Streaming Bronze (optional, proposed):** bronze.stream_<source>, only if ADR-008 is Accepted.

F. UNITY CATALOG

Governance through Unity Catalog with the following structure:

Catalog	Schemas	Purpose
dwh_dev	bronze	Raw Bronze tables from APIs
	silver	Staging/cleaned models
	gold	Analytical marts
	quarantine	Semantic rejection tables
	snapshots	SCD2 history tables
	raw_fivetran	Fivetran-landed tables (macro_watchlist_db); prefix to be confirmed in P8

Permissions follow least privilege: pipeline identities write where needed; BI consumers read; developers get minimal development access. S3 access via Unity Catalog storage credentials and external locations.

G. ORCHESTRATION

MWAA / Airflow

Amazon MWAA runs Apache Airflow as an ephemeral environment (ADR-004, D-005, D-006):

- Created/destroyed via manual GitHub Actions workflows (not on merge to main)
- Private subnets with one NAT Gateway for portfolio cost/realism balance
- DAGs and requirements.txt live in persistent S3 (persistent stack)
- Connections/variables from AWS Secrets Manager (Airflow secrets backend)

- Airflow metadata DB is ephemeral; persistent facts in S3/Databricks
- Pre-flight DAG import/parse test runs before provisioning

Daily Pipeline Flow (Conceptual)

The intended DAG flow (to be implemented in P7):

- 1. API ingestion (World Bank, FRED) -> write raw JSON to S3
- 2. Check S3 raw availability
- 3. Trigger / wait for Fivetran sync (when enabled)
- 4. Validate raw/landing availability
- 5. Load Bronze via COPY INTO (ADR-001)
- 6. dbt build (runner: MWAA -> SQL Warehouse, ADR-003)
- 7. dbt test / data quality checks
- 8. Confirm marts available

The optional streaming path (ADR-008) is not part of the daily DAG; its dbt runner is an open question.

H. DBT

Layers

- Silver: staging models for each source (world_bank, fred, macro_watchlist_db)
- Gold: dimensional models (dim_country, dim_indicator, dim_fred_series, dim_alert_rule) and fact models (fact_macro_indicator, fact_fred_observation, fact_alert_evaluation)
- Tests: not-null, unique, relationships, accepted-values, freshness/recency, custom business rules
- Snapshots: SCD2 for entities with meaningful history (e.g., watchlist_indicator.active changes)

Execution

dbt runs from MWAA execution context against Databricks SQL Warehouse (ADR-003). Connection via environment variables; no committed profiles.yml with real values. dbt-databricks adapter with SQL Warehouse HTTP path.

I. FIVETRAN

Managed ingestion path for the operational database:

- Source: Supabase PostgreSQL (macro_watchlist_db), reached through the Supabase session pooler (behavior to verify in P8)
- Connector: PostgreSQL
- Destination: Databricks (Unity Catalog) -> raw_fivetran schema (prefix to be confirmed in P8)
- Incremental: Query-Based; exact cursor (e.g. xmin) to be confirmed in P8
- Security: TLS, dedicated read-only Fivetran database user, no credentials in Git
- Optional per D-001: core platform functions without Fivetran; P8 can be formally skipped if plan lacks Databricks destination
- Supabase and Fivetran plan limits, pricing and features: to verify in P8

J. QUARANTINE (ADR-002)

Hybrid quarantine boundary accepted:

Layer	Validation Type	Failures Diverted To	Examples
Ingestion (Python/Airflow)	Structural / Schema	S3 quarantine prefix	Malformed JSON, missing top-level fields, schema mismatch

Layer	Validation Type	Failures Diverted To	Examples
Transformation (dbt)	Semantic / Business	Unity Catalog quarantine schema	Null PK, FK violation, domain values, business rules, completeness
Streaming producer (optional, ADR-008 Proposed)	Invalid messages	S3 prefix quarantine/stream/<source>/	Structure and validation rules to be defined if ADR-008 is Accepted

All rejected records retain: source_system, source_file, batch_id, rejection_timestamp, rejection_reason. No silent drops.

K. AWS SERVICES (Approved Scope)

Only the following AWS services are in scope. Others (Redshift, Glue, EMR, Athena, Kinesis, Lambda, DynamoDB, ECR, ECS, RDS) are explicitly excluded. Kinesis stays out of scope until ADR-008 is Accepted.

- S3: Raw/landing bucket (versioned, SSE-S3, prevent_destroy), DAG bucket
- MWAA: Ephemeral Airflow environment (private subnets, NAT Gateway)
- VPC/Networking: Private subnets, NAT Gateway (single AZ for portfolio), Security Groups
- Secrets Manager: API keys, Supabase DB credentials (macro-watchlist-db secret container), Fivetran credentials, Databricks tokens
- IAM: Least-privilege roles for MWAA, Databricks, Fivetran, CI/CD
- AWS Budgets: Monthly cost guardrail ($10/month) with alerts
- CloudWatch Logs: MWAA and infrastructure logging

The operational source (Supabase PostgreSQL) is hosted outside AWS and created manually, outside AWS Terraform (ADR-007).

L. DATABRICKS

- Unity Catalog: Governance layer: catalog dwh_dev, schemas bronze/silver/gold/quarantine/snapshots/raw_fivetran
- SQL Warehouse: Smallest size, short auto-stop. Used for COPY INTO (Bronze) and dbt execution (Silver/Gold)
- Bronze: Delta tables loaded via COPY INTO from S3 raw
- raw_fivetran: Fivetran direct write destination (schema prefix TBC in P8)
- Silver: dbt staging models
- Gold: dbt mart models
- Quarantine: Semantic rejection Delta tables
- Streaming job (optional, proposed): compute type (serverless vs job cluster, versus the no-cluster constraint of ADR-001/ADR-003), checkpoint location and plan type (workspace is Free Edition) are open questions in ADR-008

M. ARCHITECTURAL DECISIONS (ADRs)

ADR	Topic	Decision	Status
ADR-001	Bronze Loading	COPY INTO via Databricks SQL Warehouse	Accepted
ADR-002	Quarantine Boundary	Hybrid: ingestion structural -> S3; dbt semantic -> UC quarantine	Accepted
ADR-003	dbt Runner	dbt from MWAA execution context against SQL Warehouse	Accepted
ADR-004	MWAA Operating Model	GitHub Actions create/destroy, private subnets + NAT	Accepted
ADR-005	RDS Exception	Amazon RDS PostgreSQL as operational source for Fivetran	Superseded by ADR-007
ADR-006	Unity Catalog grants (dev)	Unity Catalog grants in dev with single principal	Accepted
ADR-007	Supabase Operational Source	Supabase PostgreSQL replaces RDS as the operational source for Fivetran (supersedes ADR-005)	Accepted
ADR-008	Streaming Ingestion Path	Optional path P8b: feed -> producer -> Kinesis -> Databricks streaming job	Proposed

ADR-008 open questions (not decided)

- Streaming source (public WebSocket/SSE feed) and BQ-004
- Databricks compute for the streaming job (serverless vs job cluster) versus the no-cluster constraints of ADR-001/ADR-003
- Checkpoint location (the raw external location is read-only)
- dbt runner for streaming tables
- Databricks plan type (workspace is Free Edition)
- Cost

N. BUSINESS QUESTIONS

BQ-001: Macro Watchlist Evolution

How have GDP growth, inflation, and unemployment evolved over time for countries on the macro watchlist?

Grain: country + indicator + year | Dimensions: country, region, indicator, year | Metrics: value, latest, absolute/percent change, trend | Range: 10 years | Sources: World Bank + macro_watchlist_db | Status: Approved

BQ-002: US Macroeconomic Trends

What are the latest trends in the selected US macroeconomic series, and how do their values change over time?

Grain: series + observation date | Dimensions: series, date, frequency, unit | Metrics: latest, previous, absolute/percent change, trend | Range: 5 years | Source: FRED | Status: Approved

BQ-003: Alert Rule Breaches

Which configured macroeconomic alert rules are currently breached, and for which countries or indicators?

Grain: country + indicator + alert rule + latest observation | Dimensions: country, region, indicator, source_system, direction, active | Metrics: latest value, threshold, difference, breach status, latest date | Range: latest | Sources: World Bank + macro_watchlist_db | Status: Approved

BQ-004: Streaming (optional)

Not defined. It must be defined before a streaming source can be selected (ADR-008 Proposed).

Data Quality Questions (DQ-001 through DQ-007)

Pipeline status, last successful batch timestamp, rows ingested per source/batch, rows quarantined per source/batch, failing dbt tests, data freshness vs SLA, source API health (response time, error rate, last success). If ADR-008 is Accepted, streaming freshness is also shown in the data-quality dashboard.

O. SECURITY

- No secrets in Git. All credentials via AWS Secrets Manager, GitHub OIDC, or runtime secret mechanisms.
- Secret management: FRED API key, Supabase database credentials (secret macro-watchlist-db, container only, no values in Git), Fivetran credentials, Databricks auth via approved mechanisms.
- Least privilege IAM: pipeline identities write only where necessary; BI consumers read-only; developers minimal.
- Fivetran: dedicated read-only database user, TLS, no credentials in Git.
- Supabase: project created manually outside AWS Terraform; connection through the session pooler; the AWS persistent stack only holds the secret container.
- AWS root account: MFA enabled. Administrative identity is non-root.
- GitHub Actions: OIDC authentication to AWS. No long-lived AWS access keys.
- Unity Catalog: storage credentials and external locations for S3 access (no ad-hoc credentials in code).

P. COST CONTROL

AWS Budget Guardrail

Parameter	Value
Budget Name	monthly-cost-guardrail
Amount	10 USD/month
Alerts (Actual)	50%, 80%, 100%
Alerts (Forecasted)	100%
Type	Alerting/monitoring guardrail (NOT a hard spending cap)

Note: This is an alerting guardrail. It does not enforce a hard stop. Afonso must create the actual AWS Budget manually before any billable apply (manual prerequisite).

Cost Awareness

- MWAA: billed while environment exists (~$0.49/hr for small); cannot be paused, only destroyed. Creation ~20-30 min, destruction ~15-20 min (both billed).
- NAT Gateway: ~$45/month per AZ (single AZ for portfolio). Primary cost driver alongside MWAA.
- Databricks SQL Warehouse: smallest size, short auto-stop (e.g., 10 min). Never leave running between sessions.
- Ephemeral MWAA model: environments created only for planned work windows; destroyed immediately after evidence collection.
- Post-destroy verification: confirm nothing billable remains via Console/Cost Explorer.
- Supabase and Fivetran costs and free-tier limits: to verify in P8.
- Optional streaming path (ADR-008): cost per window to be estimated before the ADR can be Accepted.

Q. CURRENT PHASE

P0 - Decisions and Project Bootstrap

Architectural decisions ADR-001 through ADR-004, ADR-006 and ADR-007 are Accepted. ADR-005 is Superseded by ADR-007. ADR-008 (optional streaming path) is Proposed and must stay so until Afonso accepts it. All data sources (World Bank, FRED, Supabase PostgreSQL) are Selected. Business questions (BQ-001, BQ-002, BQ-003) are Approved. P1 implementation has NOT started.

R. STATUS SUMMARY

DECIDED (Architecture & Source Choices)

- Dual ingestion paths: API (S3 -> Bronze) and Database (Fivetran -> raw_fivetran, prefix TBC in P8)
- Bronze loading: COPY INTO via SQL Warehouse (ADR-001)
- Quarantine: Hybrid boundary (ADR-002)
- dbt runner: MWAA -> SQL Warehouse (ADR-003)
- MWAA: Ephemeral, GitHub Actions create/destroy, private + NAT (ADR-004)
- Supabase PostgreSQL: operational source for Fivetran, replaces RDS (ADR-007; ADR-005 superseded)

SELECTED (Sources)

- World Bank Indicators API (no auth, page/per_page pagination)
- FRED Economic Series API (API key, limit/offset pagination)
- Supabase PostgreSQL (macro_watchlist_db) via Fivetran PostgreSQL connector

PROPOSED (not decided)

- Optional streaming path P8b (ADR-008 Proposed): streaming source, BQ-004, Databricks compute, checkpoint location, dbt runner for streaming tables, Databricks plan type and cost are open

PENDING MANUAL VERIFICATION

- Fivetran plan supports Databricks destination
- Supabase session pooler with Fivetran Query-Based sync (exact cursor); Supabase and Fivetran limits/pricing (P8)
- Databricks workspace supports storage credentials + external locations on S3 + SQL Warehouse
- AWS root MFA, non-root admin identity, AWS Budget with alerts created

NOT YET IMPLEMENTED

- Supabase project, schema macro_watchlist and seed data (manual prerequisite, P8)
- Fivetran connector configuration
- MWAA environment
- S3 buckets
- Databricks catalogs/schemas/tables
- Terraform infrastructure
- dbt project and models
- Airflow DAGs
- GitHub Actions workflows
- Optional streaming path (P8b)